THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

031/2B **PHYSICS 2B**

(Actual Practical)

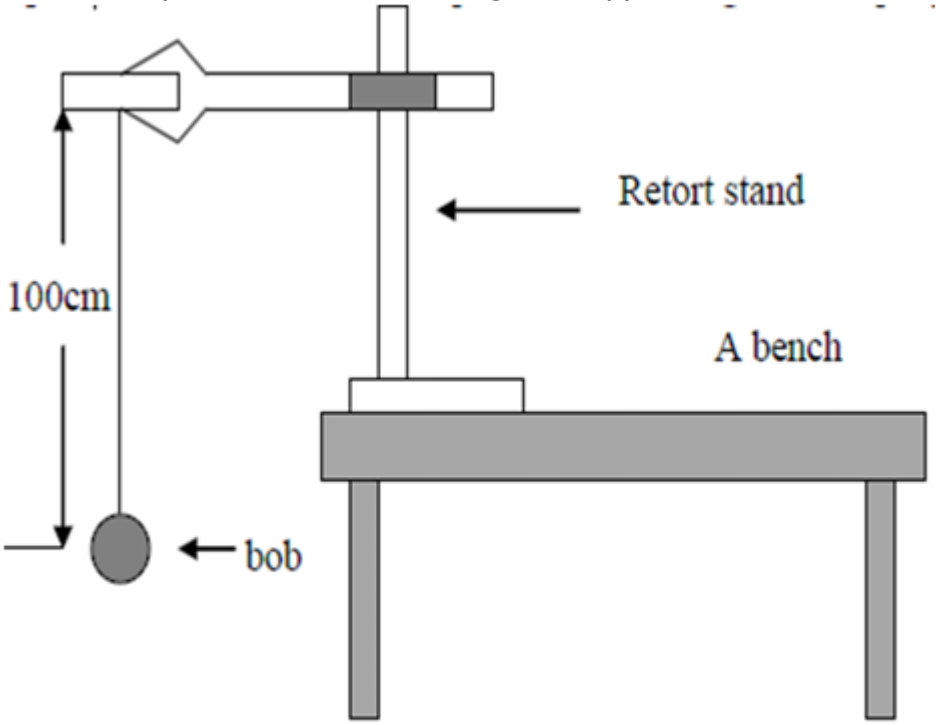
TIME: 2:30 HOURS **YEAR: 2025**

INSTRUCTIONS

1. This paper consists of **two (2)** questions.
2. Answer **all** questions.
3. Each question carries **twenty-five (25)** marks
4. Non programming calculators may be used.
5. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
6. All communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer sheets provided.

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1. The aim of this experiment is to determine the acceleration due to gravity. You are provided the following apparatus; retort stand, stop watch, cotton thread, pendulum bob, cork pads, meter rule, arrange the apparatus as shown bellow



(a) Set the apparatus given as shown in figure 1 bellow

(b) Measure length L =100cm, pull the pendulum bob aside and release it so as to swing in small amplitude

(c) Find the time t, for 20 oscillations

(d) Find the periodic time T(s) and T^2 (S^2)

(e) Repeat the procedures for L=80cm, 60cm, 40cm, and 20cm.

(f) Tabulate your results of L, t, T and T^2

(g) Plot a graph of T^2 against L.

(h) Find the slope of the graph

(i) Using the relation $T=2\pi\sqrt{L/g}$ show how you can obtain acceleration due to gravity, g

(j) Calculate the value for the acceleration due to gravity, g

(k) Write two sources of errors

2. You are provided with the following, a dry cell, resistance box, switch, an ammeter and a set of connecting wires

   (a) Connect the given electrical components in series with a switch open

   (b) draw and label clearly the circuit.

   (c) Set the resistance R= 1Ω, and then close the switch. Read and record the ammeter reading.

   (d) Repeat the procedures in (b) above by setting R = 2Ω, 3Ω, 4Ω and 5Ω

   (e) Tabulate the result including the value of 1/I

   (f) Plot a graph of R against 1/I

   (g) Find slope of the graph and vertical intercept

   (h) Determine the internal resistance and e.m.f of a dry cell

   (i) State the aim of doing experiment